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ELEVATE-LNP: Empty lipid nanoparticle enabled validation and AI-guided therapeutic evaluation for efficient formulation optimization for mRNA delivery

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ABSTRACT

Lipid nanoparticles (LNPs) have emerged as a leading platform for mRNA delivery, particularly following the success of SARS-CoV-2 mRNA vaccines. However, identifying effective LNP formulations remains time-consuming because of the large and complex multicomponent formulation space. Here, we present ELEVATE-LNP an Empty-LNP-Enabled Validation and AI-Guided Therapeutic Evaluation platform by integrating a high-throughput parallelized microfluidic preparation with physicochemical characterization and machine learning for mid-stream and downstream LNP formulation optimization when ionizable lipids are shortlisted. A broad formulation space comprising four ionizable lipids, different helper lipids, lipid ratios, and process conditions was first screened using empty LNPs to identify formulations with desirable particle size and polydispersity index (PDI). Machine-learning analysis was then used to determine the formulation and process parameters associated with empty-LNP properties. Selected candidates were subsequently evaluated following mRNA encapsulation by assessing particle size, PDI, encapsulation efficiency, cell viability, and transfection performance. The results revealed lipid-dependent relationships between formulation parameters and the physicochemical properties of empty LNPs. Six features were strongly associated with particle size, while eleven features contributed to variations in PDI. GFP expression in CFBE cells demonstrated that SM-102-based LNPs achieved higher transfection efficiency than CKK-E12-based formulations. In a separate CFTR mRNA evaluation, DOPC-containing LNPs produced higher CFTR protein expression than their DSPC-containing counterparts. By integrating empty-LNP prescreening with subsequent payload-based validation, the ELEVATE-LNP workflow reduces the number of candidates and the amount of nucleic acid required for functional testing. The platform therefore provides a practical approach for narrowing lipid-ratio and formulation-condition spaces within preselected ionizable lipid systems.

1. Introduction

Cystic fibrosis (CF) is a life-limiting genetic disorder caused by mutations in the cystic fibrosis transmembrane conductance regulator (CFTR) gene, leading to impaired ion transport, mucus accumulation, and progressive lung dysfunction [1–5]. Although small-molecule modulators have improved clinical outcomes for subsets of patients, a significant proportion of CF mutations remain unaddressed [6–8]. In this context, messenger RNA (mRNA)–based therapeutics have emerged as a promising strategy to restore functional CFTR protein expression in a mutation-agnostic manner [9]. However, the intrinsic instability of mRNA and its limited ability to cross biological barriers necessitate the development of efficient delivery systems [10,11]. Although mRNA-mediated protein restoration is transient, repeated administration can sustain therapeutic CFTR levels over time, as supported by early clinical studies such as MRT5005, highlighting the potential of mRNA delivery as a viable treatment modality for CF. [12,13]

Lipid nanoparticles (LNPs) have become the leading platform for

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mRNA delivery, owing to their ability to encapsulate nucleic acids, protect them from degradation, and facilitate intracellular delivery [14–19]. In recent years, substantial efforts have been devoted to the development of novel ionizable lipids to enhance delivery efficiency and biocompatibility [20–22]. Despite these advances, the design of LNP formulations remains largely empirical. LNP performance is governed by a complex interplay of formulation parameters, including ionizable lipids, helper lipids, cholesterol, and PEG-lipids, which collectively influence nanoparticle size, stability, and biological activity [23–28]. The high dimensionality of this formulation space, coupled with nonlinear interactions between components, makes rational optimization challenging and often reliant on trial-and-error approaches [29].

Microfluidic technologies have been widely adopted to improve the reproducibility and control of LNP synthesis, enabling precise manipulation of mixing conditions and particle formation [30,31]. Nevertheless, conventional microfluidic platforms are typically limited in throughput, restricting their ability to efficiently explore large formulation spaces. In parallel, machine learning approaches have been introduced to accelerate LNP design by modeling relationships between formulation parameters and physicochemical properties [32]. Recent studies have further demonstrated the potential of deep learning, molecular descriptors, foundation models, and multi-objective AI strategies to accelerate ionizable lipid discovery and improve LNP engineering for mRNA delivery [33–35]. For example, the AGILE platform integrates deep learning with combinatorial chemistry for rapid ionizable lipid development [33]. These studies highlight the importance of incorporating lipid structural information and molecular-level descriptors into predictive models. However, many formulation studies still rely primarily on categorical lipid identity or bulk formulation variables, and the integration of high-throughput formulation-level datasets with payload-specific biological evaluation remains limited. Furthermore, the lack of high-quality, experimentally consistent datasets continues to constrain the robustness and generalizability of ML models for LNP formulation design.

A critical but underexplored challenge lies in bridging early physicochemical optimization with payload-specific functional evaluation. Direct screening of mRNA-loaded LNPs is costly and labor-intensive because it requires nucleic acid payloads, encapsulation measurements, and cell-based assays. Empty LNPs provide a more efficient route for early-stage screening by identifying formulations with suitable size and polydispersity index (PDI). Because mRNA loading can increase particle size, selecting smaller empty LNPs may help mRNA-loaded particles remain within the commonly used sub-150 nm range for cellular delivery. Therefore, empty LNPs should be used as an upstream filtering step rather than a direct substitute for mRNA-loaded characterization.

To address these challenges, we developed ELEVATE-LNP (Empty-LNP Enabled Validation and AI-guided Therapeutic Evaluation), an integrated microfluidics–machine learning framework for high-throughput formulation optimization and biological validation, as illustrated in Fig. 1. The framework follows a stage-gated strategy to screen broadly, learn efficiently, and validate therapeutically. First, ELEVATE-LNP screens broadly by exploring a large formulation space of empty LNPs using parallelized microfluidic synthesis with MiNANO-form [36] and high-throughput DLS characterization. It then learns efficiently through machine learning–based analysis and down-selection of candidates with desirable physicochemical properties. Selected formulations are subsequently evaluated after GFP mRNA encapsulation in CF-relevant cellular models to generate payload-specific physicochemical and biological performance data, including mRNA-loaded particle size, PDI, encapsulation efficiency, cell viability, and transfection efficiency. These data are further incorporated into predictive modeling to prioritize formulations for functional mRNA delivery. Finally, the top-ranked candidates are validated therapeutically using CFTR mRNA to assess disease-relevant CFTR protein expression. Overall, this stage guided workflow enables progressive refinement from broad empty-LNP screening and model-guided selection to payload-specific evaluation and CFTR mRNA validation.

2. Materials and methods

2.1. Materials

SM-102, ALC-0315, DLin-MC3-DMA (MC3), DSPC, DOPC and DMG-

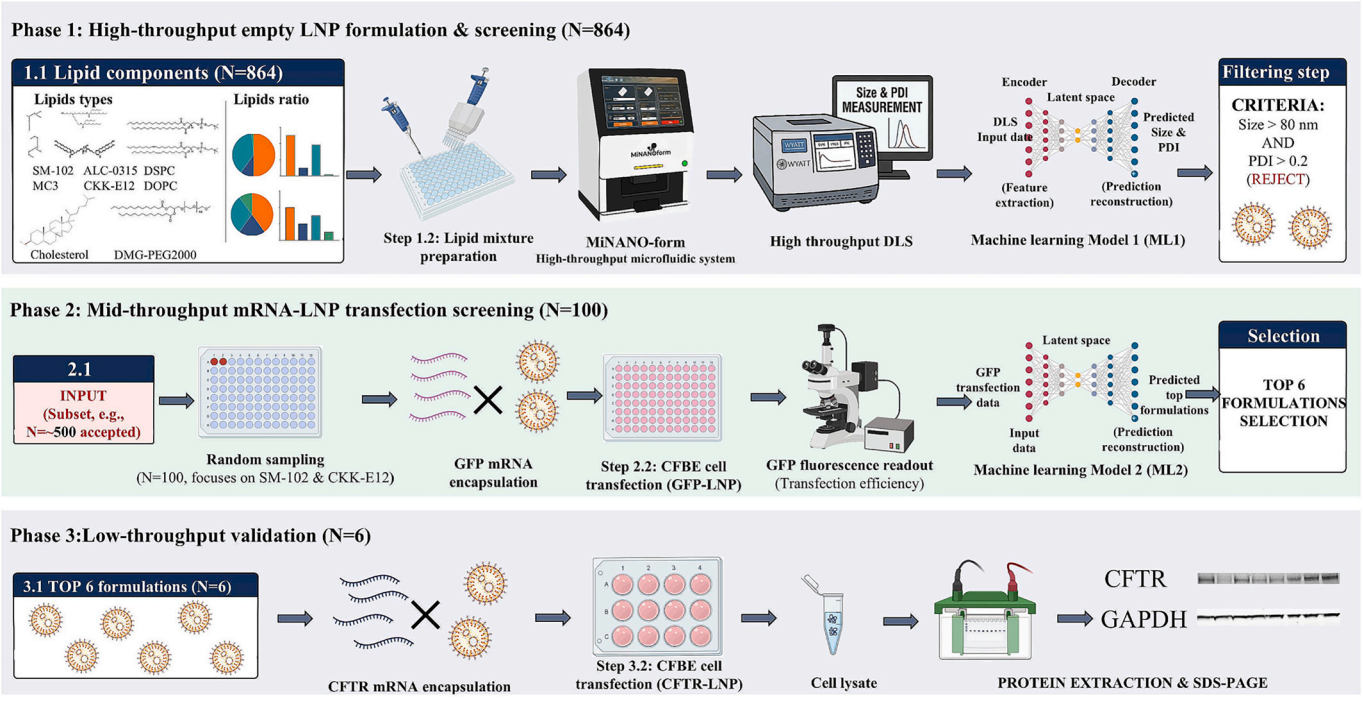


Fig. 1. ELEVATE-LNP workflow for high-throughput optimization of LNP formulations. Fig. 1 was partially created using BioRender.com and subsequently edited in Microsoft PowerPoint.

PEG2000 were sourced from Avanti Polar Lipids Inc. (Alabama, United States). CKK-E12 was purchased from MedChemExpress (MCE, USA). Cholesterol, ethanol, and sodium acetate buffer (NaOAc) were purchased from Sigma-Aldrich (Darmstadt, Germany). Enhanced Green Fluorescent Protein messenger RNA (GFP mRNA) was purchased from CATUG Biotechnology (CATUG, Basel, Switzerland). CFBE41o-human CF bronchial epithelial (CFBE) cells, phosphate buffered saline (PBS) and 1× MEM non-essential amino acid (non-EAA) were purchased from Merck. Minimum Essential Medium Eagle (MEM) was purchased from Merck Life Science. Fetal bovine serum (FBS), penicillin/streptomycin (p/s), 2mM L-Glutamine (L-Glu), Quant-it™ RiboGreen Reagent and RNA Assay Kit (RiboGreen), RIPA buffer, 7% NuPAGE Tris-Acetate protein gels, Pierce BCA Protein Assay Kit, iBright CL750 system, and Triton X-100 were ordered from Thermo Fisher Scientific (Waltham, MA). Size and PDI were measured by DynaPro ™ Plate Reader III (Waters Corporation, United States of America). The mRNA concentration and cell viability were measured by SpectraMax M3 multi-plate reader (Molecular Devices, San Jose, CA). Primary antibodies included mouse anti-human CFTR antibody (UNC-596, University of North Carolina, Chapel Hill, NC, USA) as the target protein, and mouse anti-GAPDH antibody (Abcam, Cambridge, UK) as the loading control. The secondary antibody was an HRP-conjugated anti-mouse IgG (Cell Signaling Technology, Danvers, MA, USA). An Olympus CKX41 inverted fluorescence microscope (Olympus, Tokyo, Japan) equipped with LCACHN 40×/0.55 PhP objectives, and a QImaging CCD camera, was used to visualize the expression of the reporter GFP.

2.2. High-throughput LNP formulation library construction and characterization

Four ionizable lipids (SM-102, ALC-0315, MC3, and CKK-E12) were selected, together with helper lipids (DSPC or DOPC), cholesterol, and DMG-PEG2000. The formulation design was guided by clinically validated ratios (e.g., Onpattro, Moderna) [37,38], but expanded to cover ionizable lipid concentrations ranging from 1.25 to 10 mM and systematically varied molar ratios (Table 1). In total, 864 distinct formulations were generated.

LNPs were synthesized using the parallelized microfluidic platform MiNANO-form (as shown in Fig. 1). Lipids were dissolved in ethanol (organic phase, 100 μL) and mixed with 25 mM sodium acetate buffer (NaOAc, pH 5.0; aqueous phase, 300 μL) at a flow rate ratio of 3:1 and a total flow rate of 2000 μL/min. Based on the previous study [36], increasing the total flow rate above 2000 μL/min did not result in further changes in particle size. After microfluidic mixing, 40 μL of the LNP suspension was dispensed into 96-well plates prefilled with 160 μL PBS. Size and PDI were measured using a high-throughput DynaPro™ Plate Reader III (5 s per measurement, 30 acquisitions per sample). Individual acquisitions reporting a PDI value of 0 were excluded from analysis prior to calculating the final particle size and PDI values.

2.3. AI-driven optimization of empty LNP formulations

To enable systematic exploration of the formulation space, key formulation variables including lipid types, molar ratios, and concentrations were expanded into a comprehensive feature set consisting of formulation component molecular concentrations, formulation component ratios, lipid type information, and pairwise ratio features. The original six formulation variables did not fully capture the absolute concentrations and relative proportions of all four components. Therefore, additional concentration- and ratio-derived descriptors were generated to provide a more complete physicochemical representation of each formulation. Pairwise ratio features were further included to explicitly encode the relative balance between lipid components, since normalization of individual features may reduce the representation of component-to-component proportional relationships. This feature-engineering step is used to improve physicochemical representation and support predictive model training.

Table 1
Lipid composition parameters for LNP formulation optimization.

Ionizable lipid type (ILT)	Helper lipid type (HLT)	Ionizable lipid concentration (ILC, mM)	Ionizable lipid ratio (ILR, %)	Helper lipid ratio (HLR, %)	PEG lipid ratio (PLR, %)
SM-102	DSPC	1.25	40	4	0.5
ALC-0315	DOPC	2.5	50	10	1.5
MC3		5	60	16	2.5
CKK-E12		10			

The initial dataset comprised six primary features: ionizable lipid type, helper lipid type, ionizable lipid ratio (%), helper lipid ratio (%), PEG-lipid ratio (%), and ionizable lipid concentration (mmol/mL). Eleven additional engineered features were subsequently generated, yielding a total of 17 features grouped into four categories:

- **Formulation component molecular concentrations**: ionizable lipid concentration (ILC), helper lipid concentration (HLC), PEG-lipid concentration (PLC), cholesterol concentration (CC), and total lipid concentration (TLC).
- **Formulation component ratios**: ionizable lipid ratio (ILR), helper lipid ratio (HLR), PEG-lipid ratio (PLR), and cholesterol ratio (CR).
- **Lipid type information**: ionizable lipid type (ILT) and helper lipid type (HLT).
- **Pairwise ratio features**: ionizable/helper ratio (I_H_R), ionizable/PEG ratio (I_P_R), ionizable/cholesterol ratio (I_C_R), helper/PEG ratio (H_P_R), helper/cholesterol ratio (H_C_R), and PEG/cholesterol ratio (P_C_R).

The prediction of size was formulated as a regression task, whereas PDI prediction was framed as a binary classification problem (≤ 0.2 vs. > 0.2). Candidate algorithms for size prediction included Gaussian Process Regression (GPR), Support Vector Regression (SVR), Random Forest (RF), Gradient Boosting Regression (GBR), Extreme Gradient Boosting Regression (XGBR), Ridge Regression (Ridge), and Artificial Neural Network (ANN). For PDI classification, the candidate algorithms were RF, Gradient Boosting (GB), Extreme Gradient Boosting (XGB), Gaussian Process (GP), Support Vector Machine (SVM), and Multi-Layer Perceptron (MLP). These algorithms were selected because the dataset is relatively small and the underlying relationships between LNP formulation variables and physicochemical properties are likely nonlinear. The chosen models are well suited for small-sample settings while maintaining sufficient flexibility to capture complex nonlinear input–output mappings. Potential outliers in particle size and PDI were assessed separately before model development. For each formulation, the five nearest neighbours were identified using Euclidean distance in the normalized formulation-feature space, excluding the formulation itself. The local deviation was defined as the absolute difference between the observed endpoint value and the mean value of its five nearest neighbours. For each endpoint, formulations with local deviations exceeding $Q3 + 0.75 \times IQR$ of the corresponding deviation distribution, where Q3 is the third quartile and IQR is the interquartile range, were excluded from model training. A 5×5 nested cross-validation strategy with 20 repetitions was employed for robust model selection and hyperparameter optimization. The dataset was split into five outer folds, with one-fold held out for testing and the remaining four subjected to a five-fold inner loop for hyperparameter tuning. The best parameters from the inner loop were used to train a model evaluated on the outer test fold, and this process was repeated across all outer folds. To reduce variance from a single random split, the entire procedure was repeated 20 times with different random seeds, yielding 100 outer test evaluations per model and providing a robust distribution of performance metrics.

After selecting the best-performing model for each task, SHAP-based recursive feature elimination (SHAP-RFE) was applied to refine the input space. At each iteration, one feature was removed, the model was retrained, and performance was evaluated using 10-fold cross-validation, with hyperparameters fixed to those most frequently selected during nested cross-validation. Specifically, for size prediction, features were retained if their removal increased the cross-validated mean absolute error (MAE). For PDI classification, features were retained if removing them reduced the precision of the monodisperse class (PDI $\leq$ 0.2). This process produced compact feature sets that preserved predictive accuracy while improving interpretability.

The optimized models were applied to screen a candidate design space generated by systematic sampling of lipid ratios (ionizable lipid: 40–60%, helper lipid: 4–16%, PEG-lipid: 0.5–2.5%) and total lipid concentrations (0.0025–0.01 mmol/mL). Formulations predicted to yield particle sizes below 80 nm with monodisperse PDI ($\leq$ 0.2) were shortlisted. Sobol sequence sampling was subsequently employed to maximize chemical and compositional diversity within this subset.

2.4. High-throughput optimization of GFP mRNA-loaded LNPs

A total of 100 formulations were selected from the screened formulation library for GFP mRNA encapsulation and transfection evaluation. This subset included 50 SM-102-based and 50 CKK-E12-based formulations, each combined with DSPC or DOPC. SM-102 and CKK-E12 were selected as representative ionizable lipids for the initial mRNA-loaded validation because both have strong relevance to mRNA delivery applications. CKK-E12 has been used as a reference ionizable lipid in cystic fibrosis-related mRNA delivery studies [20], while SM-102 showed favorable transfection performance in previous comparative evaluations and our preliminary tests [25,36]. The organic phase was prepared as described in Section 2.2, while the aqueous phase was prepared by diluting GFP mRNA in NaOAc to achieve an N/P ratio of 8. This N/P ratio was selected based on preliminary optimization experiments comparing N/P ratios of 6, 8, and 10, in which N/P 8 provided the best overall balance between particle size, PDI, and GFP mRNA transfection efficiency (as shown in Fig. S1). The aqueous and organic phases were loaded into inlet 1 and inlet 2 of the MiNANO-form cartridge, respectively, and mixed in the MiNANO-form system at a total flow rate of 2000 μL/min with an aqueous-to-organic phase flow rate ratio of 3:1. Following microfluidic mixing, all LNP samples were subjected to ultrafiltration to remove ethanol and exchange the buffer to PBS. The collected LNPs were diluted fivefold with PBS and further purified by ultrafiltration (10,000 ×g, 5 min, Eppendorf® centrifuge), and characterized for size and PDI using the DynaPro™ Plate Reader III. The mRNA concentration and EE were quantified using the Quant-iT™ RiboGreen assay, with EE calculated as:

$$EE\% = \frac{Total\,mRNA\,concentration - upcapsulated\,mRNA\,concentration}{Total\,mRNA\,concentration} \times 100 \quad (1)$$

For in vitro transfection, CFBE cells were seeded in 96-well plates at a density of 10,000 cells per well in MEM supplemented with 10% FBS, 1% p/s, 1% MEM non-EAA, and 1% L-glu, and incubated at 37 °C with 5% CO_2. After 24 h, cells were transfected with 600 ng GFP mRNA–loaded LNPs in a total volume of 200 μL. After 48 h, GFP expression was evaluated using an Olympus IX81 fluorescence microscope, and fluorescence intensity was semi-quantified using ImageJ Fiji software.

Cell viability was determined by replacing the culture medium with 10% AlamarBlue solution (diluted in PBS) and incubating for 1.5 h at 37 °C with 5% CO_2. Fluorescence intensity was recorded at an excitation wavelength of 560 nm and an emission wavelength of 590 nm using a SpectraMax M3 multi-plate reader (Molecular Devices, San Jose, CA). Untreated cells served as the normalization control (100% viability). Background fluorescence was subtracted from wells containing only alamarBlue™ reagent, and cell viability was calculated using the following equation:

$$\frac{\text{Fluorescence intensity of treated cells}}{\text{Fluorescence intensity of untreated cells}} \times 100\% \quad (2)$$

2.5. AI-driven optimization of GFP mRNA-loaded LNPs

To integrate physicochemical characteristics with biological performance, GFP mRNA transfection data from the 100 representative formulations (selected via Sobol sampling) were incorporated into the AI pipeline. Following log transformation of the transfection efficiency values, a regression framework was established to directly predict transfection efficiency from formulation descriptors, following the same feature engineering, nested cross-validation, and SHAP-RFE in Section 2.3. The model was developed as an exploratory tool for candidate prioritization within the experimentally sampled formulation space, rather than for resolving lipid-class-independent effects of individual formulation descriptors. Because transfection efficiency was treated as a regression endpoint, SHAP-RFE was performed using MAE as the primary performance criterion, consistent with the size-prediction task. Model predictions were extended to all formulations previously filtered by physicochemical thresholds (size <80 nm, PDI $\leq$ 0.2). The top six candidates, representing both DSPC- and DOPC-based formulations, were subsequently selected for downstream experimental validation.

2.6. CFTR mRNA encapsulation and western blot analysis

The six lead formulations were synthesized with CFTR mRNA for functional validation. CFBE cells were seeded in 12-well plates (60,000 cells/well) and treated with LNPs containing 3 μg CFTR mRNA. After 48 h, cells were lysed with RIPA buffer containing protease inhibitors. Total protein was quantified with a BCA assay, separated by 7% Tris-Acetate gels, and transferred to nitrocellulose membranes. Western blotting was performed using a mouse anti-human CFTR primary antibody (UNC-596, 1:500) and a mouse anti-GAPDH primary antibody (Abcam, 1:2000) as the loading control. The secondary antibody used for detection was an HRP-conjugated anti-mouse IgG (Cell Signaling Technology, 1:2000). Bands were visualized using Pierce™ ECL substrate and imaged with an iBright CL750 system.

2.7. Data analysis

The mean ± standard deviation (±SD) was presented for all data. For nanoparticle characterization (size and PDI), each formulation was measured in triplicate within a single preparation to account for technical variation. For cell-based experiments, each condition was tested in three wells per experiment. One-way ANOVA with Dunnett's multiple-comparison tests was used to analyze the difference for different groups using Prism 8.0 (GraphPad). A difference was considered significant if $P < 0.05$ (*$P < 0.05$, **$P < 0.01$, ***$P < 0.001$, ****$P <$ 0.0001). The complete source code and machine-learning workflow used in this study are publicly available at https://github.com/luhan611/LNP-Formulation-Screening.git.

3. Results

3.1. Construction and characterization of 864 empty LNPs

To investigate the physicochemical space of different lipid compositions, a library of 864 LNP formulations was prepared using the MiNANO-form system and analyzed using a DynaPro™ Plate Reader III (Fig. 2 and Supplementary Data 1). Fig. 2 provides an overview of the population-level size and PDI distributions grouped by ionizable lipid type, while the complete formulation-level compositions and measured physicochemical properties are provided in Supplementary Data 1.

Overall, the measured particle sizes predominantly fell within 40–100 nm, although the distribution profiles differed among ionizable lipid groups. MC3-based formulations were mainly concentrated around smaller particle sizes, whereas ALC-0315- and SM-102-based formulations showed broader population-level distributions. CKK-E12-based formulations covered a wider size range, reflecting the heterogeneity of formulation compositions within this ionizable lipid group. PDI values were largely distributed between 0.05 and 0.50, with SM-102- and MC3-based formulations showing narrower dispersity distributions than ALC-0315- and CKK-E12-based formulations. Importantly, each formulation was individually characterized and retained as a traceable prescription-level entry in Supplementary Data 1, including ionizable lipid type, helper lipid type, lipid ratios, total lipid concentration, particle size, and PDI. This prescription-level dataset provided the basis for subsequent machine learning-based modeling and formulation-feature interpretation.

3.2. Empty LNP formulation optimization based on size and PDI

3.2.1. Model selection for size and PDI prediction

Following endpoint-specific outlier removal, 824 formulations were included in the size regression and PDI classification analyses, respectively. As shown in Fig. 3a and b, regression and classification tasks revealed distinct performance differences. For size prediction, GPR, SVR, RF, GBR, and XGBR achieved similar accuracy, with average percent errors around 11.5%, MSE values of 44–48, MAE values of 5.1–5.4, and R^2 values between 0.76 and 0.78, and exhibited stable distributions across 100 nested CV runs. Ridge and ANN performed worse, with higher errors and lower R [2], the latter showing highly scattered predictions. GPR was therefore selected as the most suitable regressor, yielding the best R^2 and lowest errors. For PDI classification, RF, XGB, GB, and GP consistently achieved accuracies of 0.69–0.75, F1 scores of 0.80–0.85, and precision and recall above 0.80, with stable results across repetitions. In contrast, SVM was less consistent, and Logistic Regression performed poorly (accuracy below 0.3, near-zero precision and recall, minimal F1). RF was selected for PDI classification because it showed high accuracy and consistent precision across repeated nested CV runs.

3.2.2. SHAP-based recursive feature elimination for size and PDI prediction

SHAP analysis was employed to interpret the contribution of individual formulation features (Fig. 3c and d). A total of 17 formulation descriptors were considered in Section 2.3. Among these, four categorical variables representing ionizable lipid types (ILT-ALC-0315, ILT-SM-102, ILT-MC3, and ILT-CKK-E12) were retained as fixed factors to assess the impact of lipid chemistry on model performance. The remaining 15 continuous descriptors were subjected to SHAP analysis to evaluate their contributions to model predictions and support subsequent feature optimization.

To further refine the predictive models, a stepwise feature elimination strategy was applied. In this process, one feature was removed at a time, and the model performance was re-evaluated. If the prediction accuracy improved after eliminating a feature, it was considered redundant and excluded from the subsequent iterations. For size prediction (Fig. 3e), reducing the feature set from 17 to 6 key continuous descriptors (ILC, PLC, CC, TLC, CR, and ILR) improved performance, yielding $R^2 = 0.84$, MAE = 4.46, and MSE = 32.93, with percent error reduced to 9.93%. For PDI classification (Fig. 3f), eliminating less informative variables improved accuracy from 0.77 to 0.78, precision from 0.83 to 0.84, and recall from 0.89 to 0.90. The final optimized

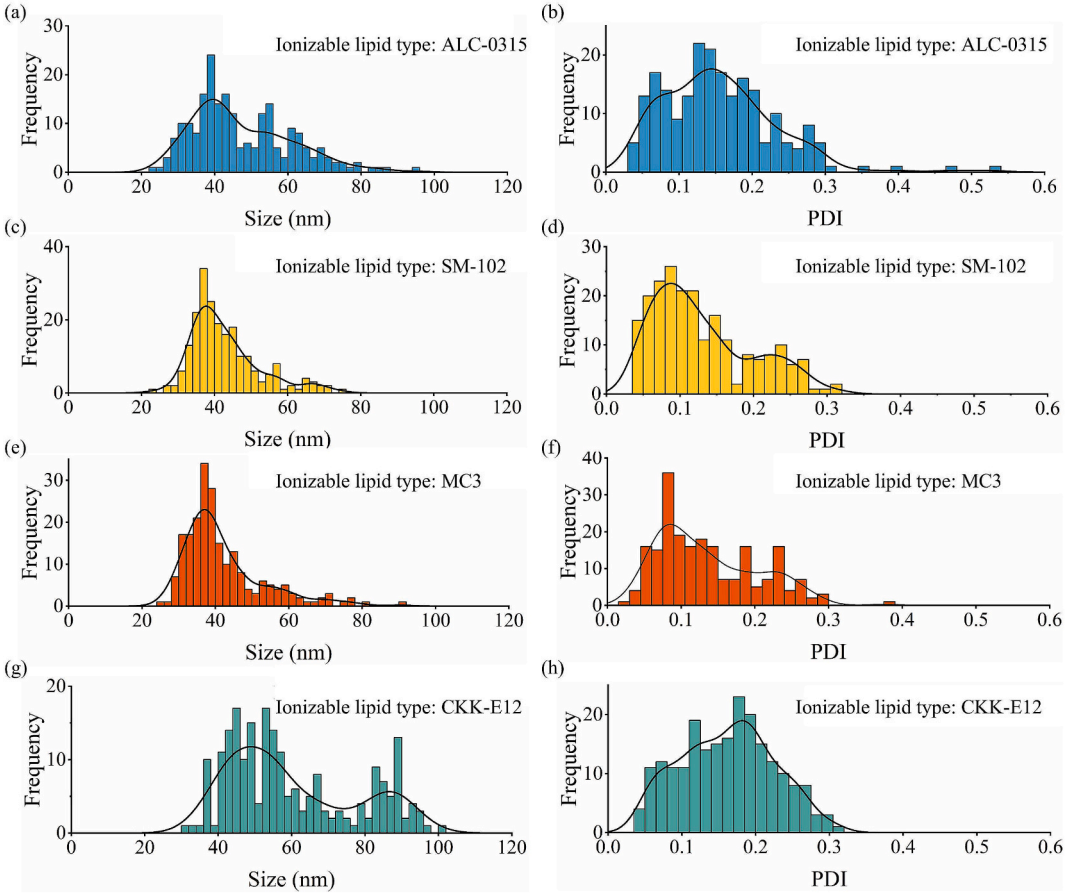


Fig. 2. Size (a, c, e, g) and PDI (b, d, f, h) distributions of 864 LNP formulations prepared with different ionizable lipids: ALC-0315 (a, b), SM-102 (c, d), MC3 (e, f), and CKK-E12 (g, h).

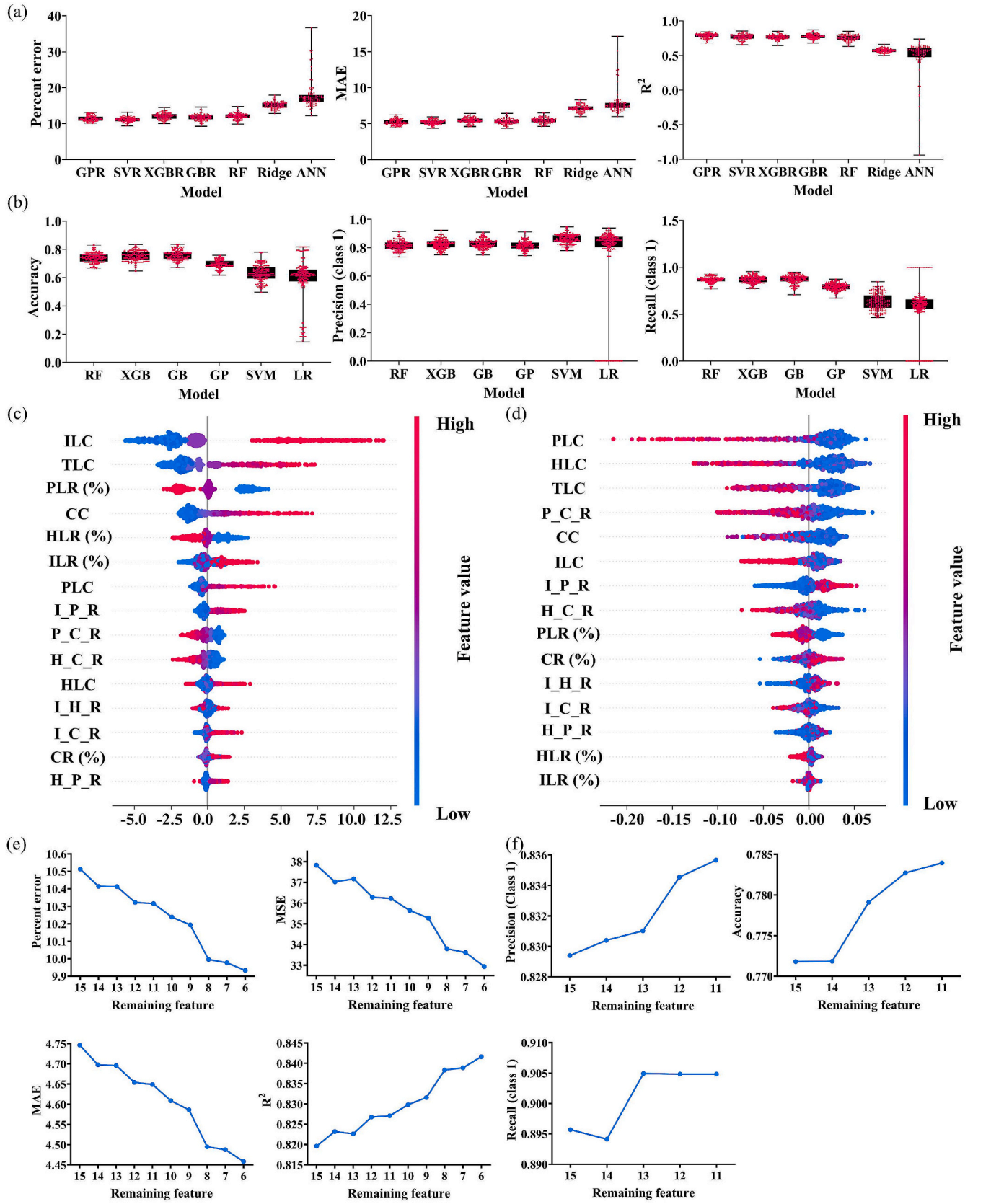


Fig. 3. Model performance, feature importance, and SHAP-RFE for LNP size and PDI prediction. (a) Percent error, MAE, and R^2 of regression models for size prediction; (b) Accuracy, precision, and recall of classification models for PDI prediction; (c) SHAP plot showing feature importance for size prediction; (d) SHAP plot showing feature importance for PDI classification; (e) SHAP-RFE improving size prediction accuracy; (f) SHAP-RFE improving PDI classification performance.

feature set included H_P_R, I_H_R, CR, PLR, I_P_R, ILC, CC, P_C_R, TLC, HLC, and PLC.

3.2.3. Feature contribution and data scaling analysis

To evaluate whether the dataset size was sufficient to support reliable model training, a learning curve analysis was first performed by progressively increasing the training set size and assessing the predictive performance of the Gaussian Process Regressor (GPR) on the size endpoint (Fig. 4b). The test MAE decreased monotonically from approximately 7.65 at 40 samples to below 4.54 at 659 samples, indicating improved predictive accuracy with increasing data availability. The most pronounced reduction in error occurred at smaller sample sizes, while the curve gradually plateaued beyond ~400 samples, suggesting diminishing returns from further data expansion. These results indicate that the current dataset provides sufficient coverage to capture the dominant formulation–property relationships, although additional data may still yield incremental improvements.

To further investigate the relative contribution of different feature groups to model performance, a systematic feature group ablation (FGA) analysis was conducted. In this procedure, the GPR model was retrained under *keep-only* configurations, where only one or a combination of predefined feature groups were retained, together with ionizable lipid type (one-hot encoded) as baseline covariates. The evaluated feature groups included concentration-based features (Conc), composition percentages (Comp), and derived ratios (Der), as well as their combinations. Model performance was assessed using MAE, MSE, and R [2], with results expressed as relative changes compared to the full feature set.

As shown in Fig. 4a, retaining only concentration features (Conc) results in model performance comparable to that of the full feature set, with minimal changes across MAE, MSE, and R [2]. In contrast, models trained exclusively on composition (Comp) or derived ratio features (Der) exhibit substantial performance degradation, with MAE increasing by more than twofold and MSE by over threefold, accompanied by a pronounced reduction in R^2 (ΔR [2] ~ -0.8). Combining feature groups (e.g., Comp + Conc or Conc + Der) partially mitigates this degradation but still does not outperform the concentration-only configuration. These findings indicate that absolute concentration features are the dominant determinants of predictive performance in size modeling, whereas ratio-based and compositional descriptors provide limited additional information when used independently. This observation is consistent with physicochemical principles, as absolute molar concentrations more directly govern nanoparticle self-assembly processes than relative composition descriptors.

FGA and learning curve analyses were not performed for the PDI classifier. As shown in Fig. S3, the model exhibits limited predictive performance for class 0, while maintaining relatively high precision for class 1 (PDI $\leq$ 0.2), which represents the primary target for formulation optimization. This enables reliable identification of low-PDI formulations, consistent with the intended application of the model. Given this task-oriented design and the imbalance in predictive performance between classes, further ablation or scaling analyses were not considered informative for the PDI model.

3.2.4. AI-driven formulation optimization

SM-102 and CKK-E12 were selected for further GFP mRNA encapsulation. To explore the formulation landscape, a broad design space was established by varying ionizable lipid ratios, helper lipid ratios, PEG-lipid ratios, and total lipid concentrations. The trained regression model for size and the classification model for PDI were then applied to evaluate all candidate formulations. Formulations predicted to meet the predefined thresholds (size <80 nm and PDI $\leq$ 0.2) were retained. To ensure chemical diversity and uniform representation within this refined

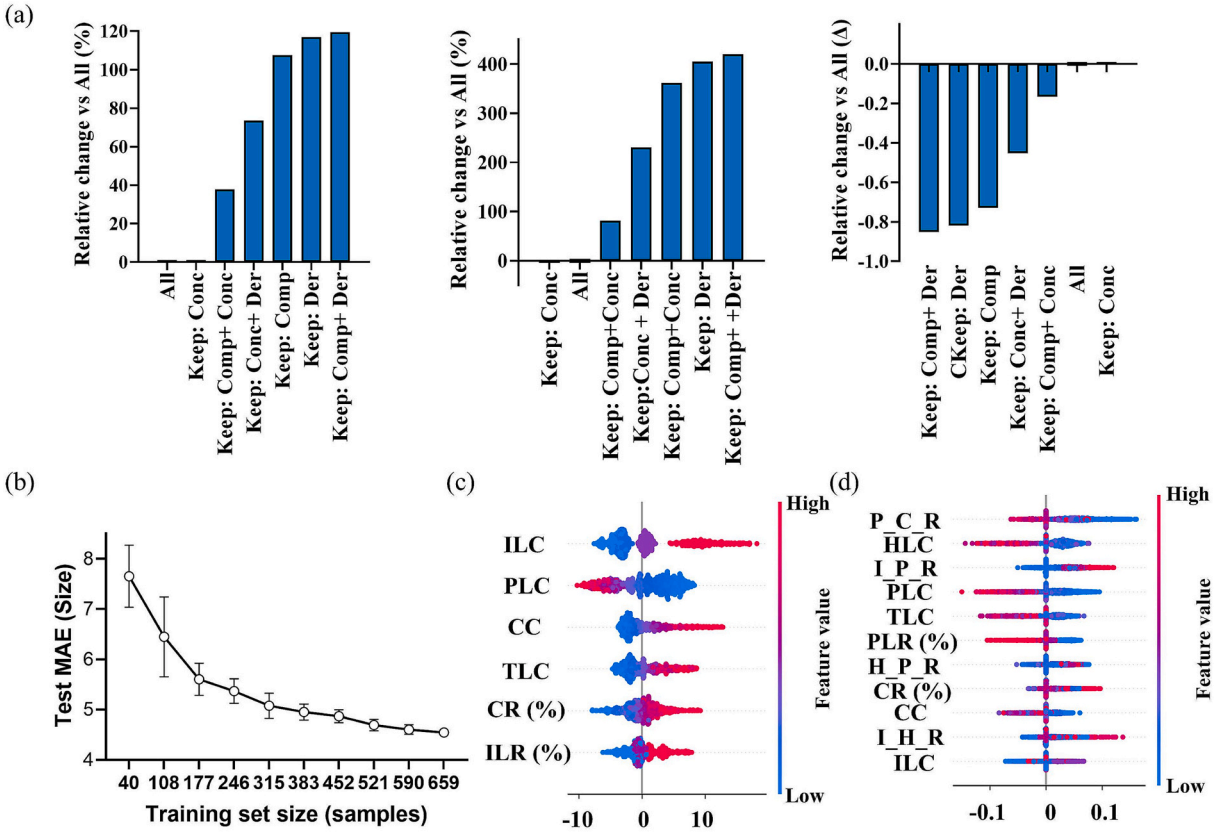


Fig. 4. (a) Feature-group ablation of the size prediction model, evaluated by MAE, MSE, and R [2]; (b) Learning curve of the size prediction model using 80% of the dataset for training and 20% as a test set; (c) SHAP plot of the refined six-feature size prediction model; (d) SHAP plot of the refined feature subset for PDI classification.

space, Sobol sequence sampling was performed, yielding a library of 100 representative formulations containing either SM-102 or CKK-E12 (Supplementary Data 2). These formulations were subsequently used for GFP mRNA encapsulation and transfection in CFBE cells to validate the AI-driven optimization framework.

To contextualize SHAP-based interpretations, pairwise Spearman correlations were first examined among formulation descriptors used in the size and PDI models (Fig. S4). The resulting correlation structure showed expected dependencies among concentration-based descriptors, molar-percentage descriptors, and pairwise ratio features. In particular, concentration-related variables such as ILC, CC, and TLC exhibited moderate to strong positive correlations, while ratio-based descriptors showed varying positive and negative associations with their constituent components. These observations indicate that multiple descriptors encode partially overlapping information about formulation composition.

Given this inherent collinearity, SHAP values were interpreted as model-specific attributions within a correlated feature space, rather than as isolated effects of independent variables. For size prediction, SHAP analysis of the refined model (Fig. 4c) identified ILC, PLC, CC, and TLC as the most influential features. Because these variables are partially correlated, their contributions were interpreted collectively, indicating that concentration-related formulation descriptors provided substantial predictive information for particle size. For PDI classification, SHAP analysis (Fig. 4d) indicated that the model used a broader set of descriptors, including ratio-based features (e.g., P_C_R, I_P_R, H_P_R) and concentration-based variables (HLC, PLC, TLC). Thus, separation of monodisperse and more heterogeneous formulations was informed by both lipid-component proportions and absolute lipid amounts, rather than by a single descriptor.

3.3. Experimental validation of model-selected formulations with GFP mRNA

To validate the AI-driven optimization, GFP mRNA was encapsulated into 100 candidate formulations and transfected into CFBE cells at an N/P ratio of 8. Prior to this screening experiment, the N/P ratio was preliminarily optimized by comparing N/P ratios of 6, 8, and 10 (Fig. S1). LNPs prepared at N/P 6 showed a larger particle size, whereas N/P 8 and N/P 10 produced smaller. Among the three tested conditions, N/P 8 generated the highest GFP fluorescence signal. Therefore, N/P 8 was selected for subsequent GFP mRNA-loaded LNP screening because it provided the best overall balance between particle size, and transfection performance. The physicochemical and biological profiles are summarized in a normalized heatmap (Fig. 5, Supplementary Data 2, and Fig. S5). Most SM-102–based LNPs exhibited particle sizes below 150 nm with PDI values under 0.2, whereas approximately 14% of CKK-E12–based formulations exceeded 150 nm. Encapsulation efficiency remained consistently high (>80%) across all formulations.

Marked differences in biological performance were observed across formulations. SM-102–based LNPs generally produced strong GFP expression, with several SM-102–DOPC formulations reaching fluorescence intensities above 2 × 10 [8], while SM-102–DSPC formulations showed moderate but consistent expression levels. In contrast, CKK-E12–based formulations exhibited fluorescence signals close to baseline. Cell viability analysis showed that SM-102 formulations largely

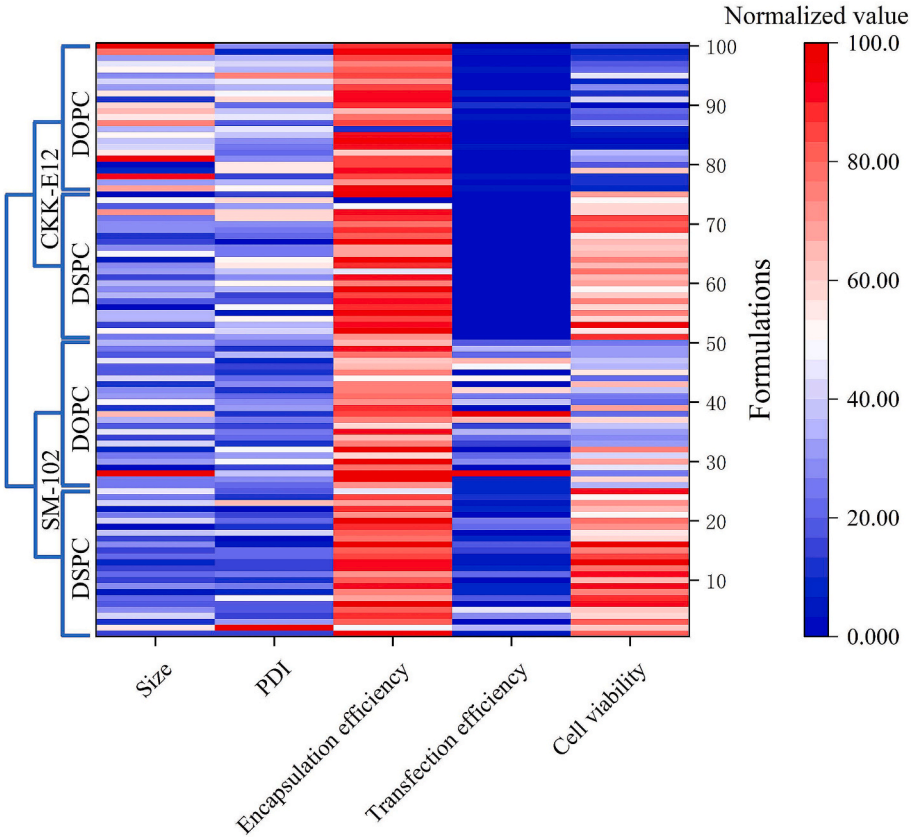


Fig. 5. Heatmap profiling of the physicochemical properties and biological performance of 100 model-selected LNP formulations. The heatmap simultaneously visualizes five evaluated parameters: particle size, PDI, EE, transfection efficiency (GFP fluorescence intensity), and cell viability across all candidate LNPs. To facilitate multi-parametric comparison within a single visual framework, the raw data for each column were min-max normalized to a relative scale of 0 to 100. The colour gradient transitions from blue (relative minimum, 0) to red (relative maximum, 100) for each specific parameter. Note that for Size and PDI, red indicates larger particle sizes or higher heterogeneity, whereas for EE, Transfection efficiency, and Cell viability, red indicates higher delivery performance or cell survival. Left-side row annotations denote the specific structural lipid and helper lipid compositions (CKK-E12 vs. SM-102, and DOPC vs. DSPC) for each clustered group.

maintained high viability (>80%), although DOPC-containing formulations displayed a slight reduction compared with DSPC-based counterparts. For CKK-E12 systems, DSPC-based LNPs preserved higher viability (>85%), whereas several DOPC-containing formulations reduced viability to ~70–80%.

Overall, SM-102–based formulations demonstrated substantially higher transfection efficiency than CKK-E12–based systems. DOPC-containing LNPs yielded stronger GFP expression than DSPC-based counterparts, accompanied by a modest decrease in cell viability.

3.4. GFP mRNA-loaded LNP formulation optimization based on transfection efficiency

GFP mRNA transfection data were generated from 100 LNP formulations comprising two ionizable lipid systems, SM-102 and CKK-E12, with 50 formulations for each system. These data were used to train regression models for transfection-efficiency prediction. Model performance was compared across multiple regression algorithms (Fig. 6a). GBR and RF showed the lowest percent errors (~4.9–5.0%) and MAE values (~0.80), while GPR gave the highest R^2 (≈0.88) with a comparable percent error (~5.4%). SVR and XGBR showed intermediate performance, whereas Ridge regression gave the weakest performance (percent error ~ 8.4%, $R^2 \approx 0.75$). GPR was selected for subsequent analysis because it provided the highest R^2 while maintaining a similar error level to the best-performing ensemble models.

SHAP analysis of the full-feature GPR model was used to rank the contributions of formulation descriptors to transfection-efficiency predictions within the combined formulation space (Fig. 6b). SHAP-RFE was then performed by removing lower-ranked descriptors and re-evaluating model performance after each removal step. This reduced the feature set to seven descriptors (Fig. 6c) and improved model performance to $R^2 \approx 0.92$ with percent error below 5%. Feature-feature

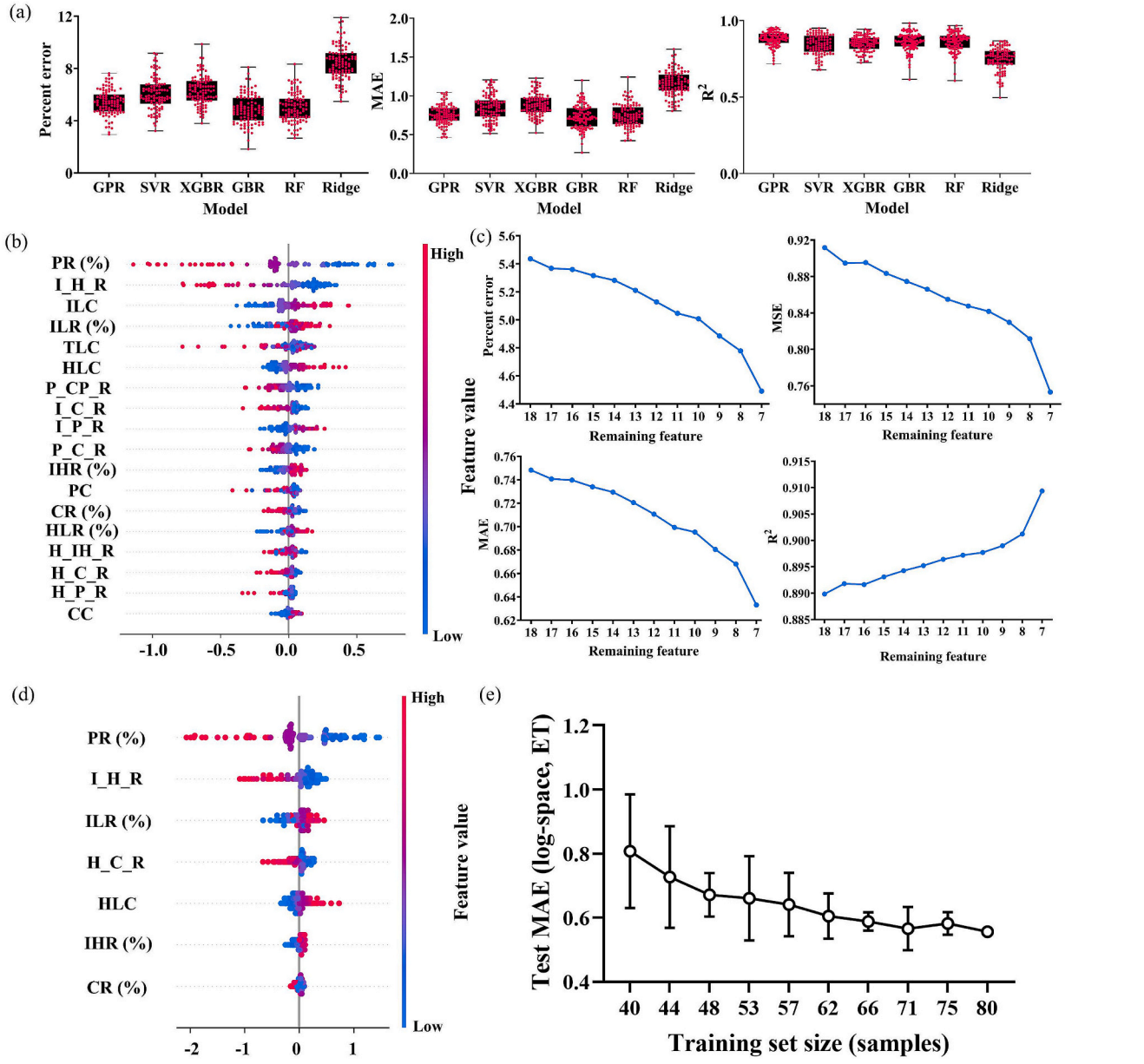


Fig. 6. Model refinement by integrating GFP mRNA transfection data. (a) Performance comparison of regression models using biological data; (b) SHAP summary plot showing feature importance for the full model (higher feature values are shown in red, lower in blue); (c) SHAP-RFE improving model accuracy; (d) SHAP plot of the refined model highlighting the seven key predictors; (e) Learning curve of the transfection efficiency prediction model using 80% of the dataset for training and 20% as a test set.

correlation analysis was performed on the GFP mRNA-loaded LNP dataset to guide interpretation of the refined SHAP results (Fig. S6). As in the empty LNP dataset, several concentration-, molar-percentage-, and ratio-derived descriptors were correlated. Therefore, the refined SHAP results were interpreted at the level of descriptor groups rather than as isolated effects of individual variables. In the seven-feature model (Fig. 6d), the retained descriptors represented PEG-lipid ratio, ionizable/helper lipid balance, and lipid concentration. PR% showed the clearest directional pattern, with higher values associated with lower predicted transfection efficiency. This negative association is consistent with the reported surface-shielding and steric-stabilizing effects of PEG-lipids, which may reduce cellular uptake and membrane interactions [39,40]. In contrast, ionizable lipid-associated descriptors, including I_H_R and ILR (%), showed positive contributions to model-predicted transfection efficiency, consistent with previous reports on the relevance of ionizable lipid content and ionizable/helper lipid balance in mRNA delivery [41–43]. The remaining retained descriptors included ionizable/helper lipid ratio-related features and concentration-derived features, indicating that the model incorporated information from both relative composition and absolute lipid amount when predicting GFP expression. Given the distinct GFP-expression distributions of the SM-102 and CKK-E12 formulations, the retained features were interpreted as predictive descriptors within the combined formulation space rather than as independent determinants of transfection efficiency. Model learning behavior was evaluated using a learning curve in log-transformed space (Fig. 6e). The total dataset size was gradually increased from 50 to 100 formulations. With a fixed 80/20 train-test split, this corresponded to training set sizes from 40 to 80 samples. Test MAE decreased from approximately 1.0 to ~0.90 as the training set size increased, with little additional improvement at larger sample sizes. At larger training sizes, the test MAE approached a plateau, suggesting diminishing improvement for the combined GPR model under the current dataset structure.

The trained model was then used to screen formulations within the SM-102 design space. This focused screening step was performed because SM-102 formulations showed a wider range of GFP expression than CKK-E12 formulations in the experimental dataset, whereas CKK-E12 formulations had consistently low transfection signals. Candidate formulations were first constrained by the predefined physicochemical criteria used in the empty LNP screen (size <80 nm and PDI $\leq$ 0.2). Among the remaining SM-102 candidates, six formulations, including three DSPC-based and three DOPC-based formulations, were selected based on predicted GFP expression for experimental validation (Table S2).

3.5. Identification of lead formulations for CFTR mRNA delivery

The six selected SM-102-based LNP formulations were next evaluated for CFTR mRNA delivery in CFBE cells (formulations were shown in Table. S2). Their physicochemical properties are shown in Fig. 7a, b. All

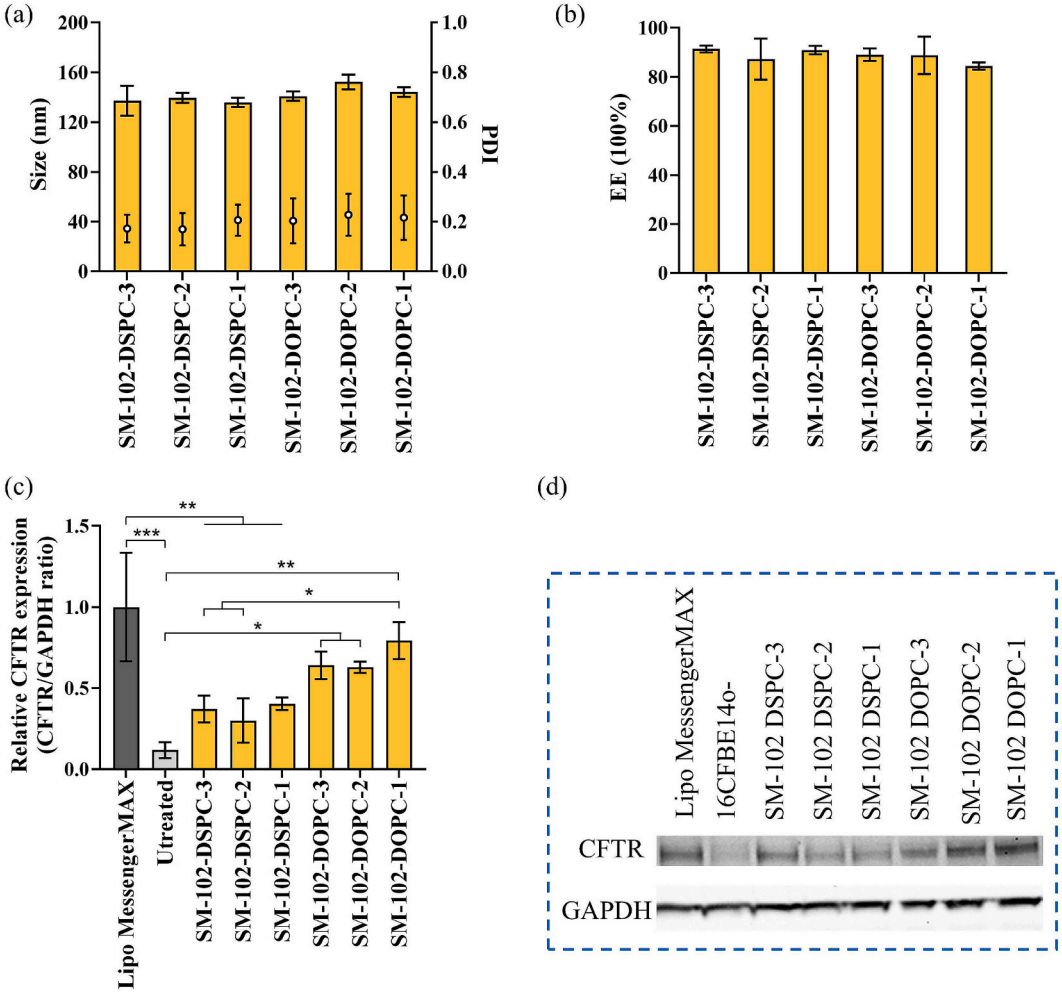


Fig. 7. Characterization and functional validation of CFTR mRNA–loaded LNPs. (a) The size and PDI of six lead SM-102 formulations; (b) EE of the same formulations; (c) Relative CFTR protein expression in CFBE cells normalized to Lipofectamine MessengerMAX Reagent (Lipo MessengerMAX); (d) Western blot showing CFTR protein expression in CFBE cells 48 h post-transfection; GAPDH served as the loading control. Lipo MessengerMAX was used as positive control.

formulations exhibited size below 150 nm with narrow dispersity (PDI ≤ 0.2) (Fig. 7a). EE was uniformly high, exceeding 80% across all formulations (Fig. 7b), confirming their suitability for mRNA loading. Functional performance was further assessed by Western blot analysis (Fig. 7d). All six formulations induced detectable CFTR protein expression. Notably, DOPC-containing LNPs generally produced stronger CFTR bands than their DSPC-containing counterparts. Densitometric quantification of the Western blot bands using ImageJ further supported this trend (Fig. 7c), with SM-102-DOPC-1 showing the highest CFTR expression among the tested LNP formulations. Although the commercial Lipo MessengerMAX control exhibited the highest overall CFTR expression, its expression level was not significantly different from that of the best-performing DOPC-containing formulation. These results are broadly consistent with the model-guided prediction that DOPC-containing formulations would show stronger CFTR mRNA delivery than DSPC-containing formulations, supporting the potential utility of the screening model for formulation prioritization.

4. Discussion

In this study, we introduce ELEVATE-LNP, a stage guided platform for mRNA–LNP formulation optimization. As summarized in Fig. 8, ELEVATE-LNP organizes formulation development into three interconnected stages: broad physicochemical space mapping using empty LNPs, payload-specific functional screening using mRNA-loaded LNPs, and disease-relevant validation using CFTR mRNA. The platform is primarily intended for midstream and downstream optimization after promising ionizable lipids have been shortlisted, allowing lipid compositions and molar ratios to be systematically refined. In the first stage, 864 empty LNP formulations containing four ionizable lipids, ALC-0315, SM-102, MC3, and CKK-E12, were prepared using a high-throughput microfluidic platform and characterized by high-throughput DLS. The resulting formulation-level dataset was used to train machine learning models for empty-LNP size and PDI prediction. These models were then applied to an expanded virtual formulation space comprising 211,344 formulations across the four ionizable lipid systems, among which 46,004 formulations met the predefined physicochemical criteria of particle size <80 nm and PDI ≤ 0.2. Because the subsequent mRNA-loaded screening focused on SM-102 and CKK-E12, 100,672 virtual formulations from these two lipid systems were further considered. Among them, 26,450 formulations satisfied the physicochemical selection criteria, including 20,884 SM-102-based formulations and 5566 CKK-E12-based formulations. To obtain an experimentally manageable and representative subset for payload-specific evaluation, 100 formulations were selected from this eligible candidate pool using Sobol sequence-based sampling, with 50 formulations from each ionizable lipid system. These formulations were then loaded with GFP mRNA and evaluated for particle size, PDI, encapsulation efficiency, cell viability, and transfection efficiency. Since SM-102-based formulations showed superior GFP mRNA transfection performance under the tested conditions, six SM-102-based candidates, including three DSPC-containing and three DOPC-containing formulations, were further selected for CFTR mRNA loading and Western blot validation.

The predicted formulation landscapes further illustrate the complexity of LNP formulation design. Across the four ionizable lipid systems, particle size and PDI exhibited nonlinear and lipid-dependent responses to variations in formulation composition, as shown in Fig. 9a and b. These findings suggest that LNP assembly may not be adequately described using conventional one-factor-at-a-time optimization, because comparable changes in lipid ratio or total lipid concentration can produce distinct physicochemical outcomes depending on the chemistry of the ionizable lipid. Consequently, formulation–property relationships identified for one lipid system may not be directly transferable to another. In this context, the ELEVATE-LNP microfluidics–machine learning framework provides a practical approach for mapping multidimensional formulation spaces and excluding candidates with unfavorable particle size or PDI before resource-intensive mRNA encapsulation and biological evaluation. Following mRNA loading, the particle-size distribution generally shifted toward larger values compared with the corresponding empty LNPs (Fig. 9c). The increase in particle size, defined as ΔS, ranged from approximately 0 to 120 nm, with a mean increase of approximately 50 nm, while most formulations exhibited increases below approximately 70 nm (Fig. 9d). These results provide an empirical reference for defining initial empty-LNP size thresholds during prescreening. Nevertheless, because the magnitude of the size increase remained formulation-dependent, ΔS should be regarded as a preliminary screening guide rather than a direct prediction rule. Experimental characterization of mRNA-loaded LNPs therefore remains essential for payload-specific formulation optimization.

The ELEVATE-LNP framework is particularly suited to midstream and downstream LNP optimization after promising ionizable lipids have

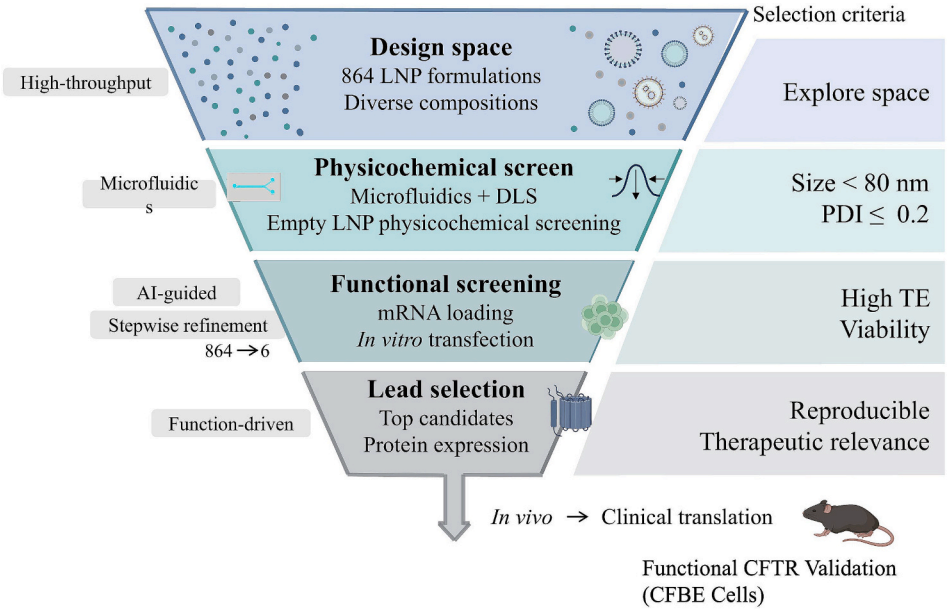


Fig. 8. ELEVATE-LNP stage guided framework for mRNA LNP formulation discovery.

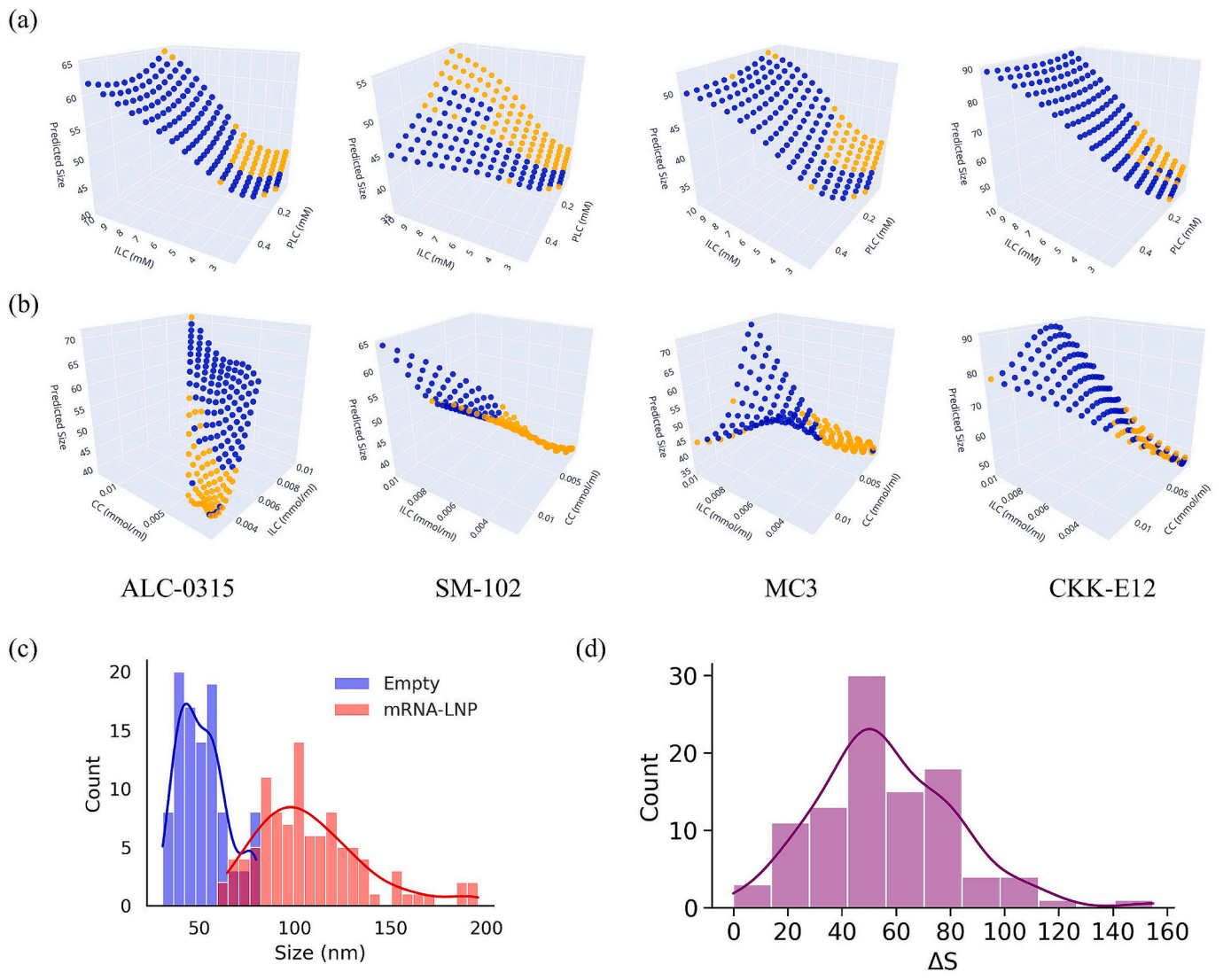


Fig. 9. Formulation landscape and size evolution upon mRNA loading. 3D visualization of predicted LNP size and PDI class for ALC-0315, SM-102, MC3, and CKK-E12 formulations: (a) HLR = 10%, PLR = 1.5%; (b) ILR = 50%, HLR = 10%. Colours indicate PDI class (blue: PDI $\leq$ 0.2; orange: PDI > 0.2); (c) Size distributions of empty and mRNA-loaded LNPs, showing a systematic increase upon encapsulation; (d) Distribution of size change (ΔS = size of mRNA-loaded- size of predicted empty size), indicating a consistent shift across formulations.

been shortlisted. During early-stage ionizable lipid development, a small-scale preliminary transfection assay may first be used to identify lipid classes with sufficient biological activity in the target cell type, while candidates with minimal transfection potential can be excluded before entering a larger formulation-screening workflow. For shortlisted ionizable lipids, ELEVATE-LNP can then be applied to systematically refine lipid composition, molar ratios, and formulation conditions. Empty-LNP screening first identifies regions of the formulation space with suitable particle size and PDI, after which mRNA encapsulation and functional evaluation are performed on a more focused candidate set. By concentrating experimental resources on preselected ionizable lipid systems, this staged approach can reduce mRNA consumption, limit the number of cell-based assays required, and improve the efficiency of lead formulation identification.

Several limitations should be addressed in future work. First, although the current study included four ionizable lipids in the empty-LNP library, the mRNA-loaded functional screening was focused on SM-102 and CKK-E12. Because SM-102-based formulations showed higher and more variable GFP expression, whereas CKK-E12-based formulations exhibited limited transfection performance under the tested conditions, the combined model may partly reflect lipid-class-associated differences. This dataset structure limits the ability to distinguish lipid-class effects from continuous formulation effects within each class and therefore constrains the mechanistic interpretation of the SHAP-RFE results. Larger lipid-specific datasets with broader within-class response distributions, together with evaluation of mRNA-loaded formulations containing additional ionizable lipids such as ALC-0315 and MC3, will be needed to resolve these effects and improve the generalizability of the workflow. Second, the present biological evaluation was performed mainly in vitro using CFBE cells. Future studies should include more physiologically relevant airway models and in vivo validation to assess biodistribution, tissue-specific delivery, safety, and therapeutic relevance. Disease-relevant models such as CF mouse models or reporter ferret models could provide important platforms for evaluating pulmonary mRNA delivery and CFTR expression. Finally, integration with automated formulation preparation and high-throughput microfluidic production systems will further improve the scalability and reproducibility of this workflow by reducing manual handling, operator-dependent variability, and experimental error.

5. Conclusion

In conclusion, this study presents ELEVATE-LNP, a stage guided framework for midstream and downstream mRNA–LNP formulation optimization. Empty-LNP characterization is used as an initial physicochemical filtering step, rather than as a substitute for direct evaluation of mRNA-loaded LNPs. By integrating high-throughput microfluidic screening, AI-guided candidate down-selection, ELEVATE-LNP enables broad exploration of formulation design spaces while significantly reducing the experimental burden associated with mRNA encapsulation and cell-based testing. In the disease-relevant CFTR mRNA validation stage, the six selected formulations showed particle sizes below 160 nm and PDI values below 0.2. Western blot analysis further showed detectable CFTR protein expression for the selected formulations, and several formulations achieved expression levels that were not statistically different from Lipofectamine MessengerMAX under the tested conditions. Overall, ELEVATE-LNP provides a practical midstream and downstream optimization strategy for refining lipid compositions, formulation ratios, and process conditions within shortlisted ionizable lipid systems. Further validation using newly developed ionizable lipids, automated workflows, and advanced airway models will be required to assess its broader scalability and translational relevance.

CRediT authorship contribution statement

Mingzhi Yu: Writing – original draft, Methodology, Data curation. **Luhan Wang:** Writing – original draft, Methodology, Data curation. **Timothy Elliott:** Methodology. **Xianqing Wang:** Methodology. **Allen Mathew:** Methodology. **Bei Qiu:** Methodology. **Liang Yao:** Methodology. **Wenxin Wang:** Funding acquisition, Writing – review & editing. **Nan Zhang:** Writing – review & editing, Supervision, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Nan Zhang reports financial support was provided by Enterprise Ireland. Nan Zhang reports financial support was provided by Research Ireland. Mingzhi Yu reports financial support was provided by China Scholarship Council. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2026.180282.

Data availability

Data will be made available on request.

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